Adaptive-depth halting as a per-perturbation signature: a reproducible, effect-size-independent, cell-type-specific property recovered from single-endpoint Perturb-seq

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Abstract

Most Perturb-seq experiments measure a single endpoint — control versus perturbed — yet the biology underneath ranges from direct target effects to feedback, compensation, and context-specific regulatory cascades. We ask whether an adaptive-depth model's *halting behaviour* — how much iterative refinement it needs to predict a perturbation's endpoint response — is a reproducible, effect-size-independent property of each perturbation, and whether it carries information beyond response magnitude. Halting depth is treated throughout as a *computational proxy for response complexity*, explicitly not biological time or causal depth, both of which a single endpoint cannot recover. Grafting learned halting (ACT/PonderNet) onto diffusion (scDiff), graph-message-passing (TxPert), and purpose-built refiners all failed in one of two ways: the halt depth collapsed to a pure effect-size proxy, or it was irreproducible across seeds. This identifiability wall motivated a magnitude-free, oracle-supervised scheme built on a *frozen* pretrained perturbation foundation model (ARC Institute STATE, `ST-SE-Replogle`), whose eight transformer layers are read as eight refinement rounds. A learnable refinement token drives a LayerNorm-stabilised halt head trained against a per-perturbation oracle stopping round; the expected halting depth $\mathbb{E}[N]$ is a stop-mass-weighted mixture over per-round decoded predictions. On K562 (968 perturbations), $\mathbb{E}[N]$ is reproducible (split-half $\rho = 0.76$), effect-size-independent (partial $\rho = -0.04$ controlling #DE and cell count), and recovers its oracle target ($\rho = 0.61$). Across four Replogle lines the learned-$\mathbb{E}[N]$ split-half reproducibility averages $\rho = 0.80$ — below the $\rho \approx 0.97$ effect-size noise ceiling — but the depth ranking does *not* port across cell types (cross-line $\rho = 0.14$; model-free ceiling $\rho = 0.011$) or across donors of primary $CD4^+$ T cells in the resting state (single-cell cross-donor $\rho \approx 0$), while porting at the stimulated endpoint where donors converge ($\rho = 0.49$). Depth is independent of knockdown strength and target abundance in all four lines, and its sign relative to response magnitude is set by the cellular context. Applied to drug-target discovery, $\mathbb{E}[N]$ does not sharpen target-class identification beyond response and network similarity (three of four lines negative). We conclude that adaptive-depth halting is a reproducible, effect-independent, cell-type-specific property of a perturbation's response recoverable from single-endpoint data, but is neither portable across contexts nor a value-add for target prioritisation — a set of rigorous negative and positive results that together delimit what halting-based signatures can and cannot claim.

1 Introduction

A genetic or chemical perturbation experiment in single cells almost always reports a single comparison: the transcriptome of perturbed cells against unperturbed controls. This control-versus-perturbed endpoint is the quantity that perturbation-response models are trained to predict, and the quantity on which they are benchmarked. Yet the biology that produces a given endpoint is heterogeneous. A knockdown of a structural ribosomal protein produces a large, immediate, and largely direct transcriptional shift; a knockdown of a signalling regulator may produce a comparable endpoint only after feedback, compensation, and a cascade of secondary regulatory events. The endpoint alone does not distinguish these regimes — two perturbations of very different mechanistic "complexity" can land at responses of identical magnitude.

We ask whether a *computational* correlate of that complexity can be recovered from the endpoint data we already have. Specifically: if a model predicts a perturbation's response by iterative refinement, how much refinement does each perturbation require, is that requirement a stable property of the perturbation, and does it carry information the response magnitude does not? We operationalise "amount of refinement" as the halting behaviour of an adaptive-depth model in the sense of Adaptive Computation Time (ACT) [1] and PonderNet [2]: a model that can repeat a refinement step a variable number of times, with a learned head deciding when to stop, yields a per-perturbation expected halting depth $\mathbb{E}[N]$.

What the signature is, and is not. We are deliberate about interpretation. Halting depth is a *computational proxy for response complexity* — the number of refinement rounds the model needs to best capture a perturbation's response direction. It is **not** biological time (the data are a single endpoint, not a time course), and it is **not** causal cascade length (a single endpoint cannot identify the causal graph). Any biological reading is strictly at the level of "this perturbation's response is captured with more or fewer refinement steps by this model," never "this perturbation acts through a longer causal chain."

Falsification-first framing. The central scientific risk is that halting depth is merely a re-encoding of effect size — larger responses simply requiring more refinement — in which case the signature is redundant and the exercise adds nothing to a norm of the predicted Δ. We therefore adopt the thesis's own test order as a gate rather than a menu: (i) **reproducibility** — is depth stable across independent samples, seeds, and splits; (ii) **non-redundancy** — does depth survive regressing out effect size, number of differentially-expressed (DE) genes, and cell count; and only then (iii) **interpretation**. If depth fails (i) or collapses under (ii), that negative result is the finding, and we report it as such. This stance — that a clean negative is a publishable result — is the correct one for a subfield in which benchmark critiques have repeatedly shown that simple baselines match elaborate models [7, 8, 9].

Contributions. This paper reports, in narrative order:

1. A code-grounded catalog of 43 perturbation-response models (2024–2026) scored for their ability to host a learned-halting refinement loop, establishing where adaptive-depth halting can even be grafted (Section 2).
2. An *identifiability wall*: learned halting grafted onto scDiff, TxPert, and purpose-built refiners collapses to an effect-size proxy or fails to reproduce. We characterise exactly which design choices cause which failure, and show the wall is the reason a naive learned gate cannot recover a per-perturbation depth from single-endpoint data (Section 3).
3. A working method — **oracle-supervised adaptive-depth halting on a frozen STATE foundation model** — that clears the reproducibility and non-redundancy gates (Section 4), with a headline K562 result of split-half $\rho = 0.76$, partial $\rho(\text{effect}) = -0.04$, and oracle recovery $\rho = 0.61$ (Section 5.1).

4. Evidence that the depth signature is **cell-type/context-specific**: reproducible within each of four cell lines (mean split-half ρ= 0.80) and within a CD4$^+$ T-cell donor (ρ= 0.62–0.75 at single-cell resolution) but not portable across lines (ρ= 0.14) or across donors in the resting/early-activation state (cross-donor $\rho \approx 0$), with a three-tier decay from same-model to cross-context reproducibility (Sections 5.2–5.3, 5.8).
5. A biological characterisation across all four lines showing depth is independent of knockdown strength and target abundance, relates to response complexity with a *cell-type-specific sign*, and is effect-size-independent once #DE and cell count are controlled (Section 5.4).
6. A negative application result: adaptive-depth halting does not sharpen drug-target-class identification beyond response and STRING-network similarity (three of four lines negative; Section 5.6).
7. A second application to primary human CD4$^+$ T-cell CRISPRi at single-cell resolution (Section 5.8): the depth signature is reproducible within a donor (split-half ρ= 0.62–0.75, rising with stimulation), donor-specific except at the stimulated 48 h endpoint (cross-donor ρ from -0.11 to 0.49), and at that endpoint separates resting-sparing inflammatory-program suppressors from damaging and generic perturbations (Cliff's $\delta = -0.16$ to -0.24, $p < 10^{-7}$).

The overall verdict is a triad of positive properties — reproducible, effect-independent, cell-type-specific — delimited by a set of equally important negatives: learned halting is unidentifiable from single endpoints without oracle scaffolding, the signature does not port across contexts, and it does not help prioritise drug targets. We regard the negatives as the core scientific contribution.

2 Related work: a graftability catalog of perturbation models

The thesis rests on an engineering precondition: there must exist a model whose forward pass contains a *semantically meaningful* iterative core over which halting can be defined. To establish where such a core exists, we compiled a code-grounded review of 43 perturbation-response models published 2024–2026, spanning five family tracks (Table 1), and read each repository to locate the exact insertion point (if any) for a learned-halting refinement loop.

What ACT requires. A learned-halting loop needs three things in the forward pass: an *unrollable core* whose repetition is semantically meaningful (running it again refines the same quantity toward a target); a *halting head* that reads the running state and emits a stop probability; and a *ponder cost* penalising expected step count so the step count becomes an informative learned quantity. The "semantically meaningful" clause disqualifies most models: a feed-forward decoder's layer 3 is not a better version of layer 2's output, so "run it more" is undefined. On this criterion, 11 of 43 models expose a graftable iterative core (tier HIGH or MEDIUM–HIGH), while 20 rank LOW or LOW–MEDIUM — adding halting to the latter would mean inventing an iterative core, not grafting a head.

Two model classes pass cleanly. **Generative transport** models (diffusion, flow-matching, neural optimal transport) have the cleanest semantics: each denoising or ODE step provably moves the sample toward the target, so "more steps $\rightarrow$ closer" is exactly what ACT wants, with no confound. **Graph message-passing** models (GEARS, TxPert, PDGrapher) also expose a loop, but "more hops" conflates *graph distance* with *response complexity* — a confound the thesis must handle. Figure 1 places the candidate hosts on a family$\times$graftability grid.

Table 1: Perturbation-model catalog (2024–2026), $n = 43$ models. Of these, 38 had a verified-active public repository with model code; 3 had a repository without model code and 2 were not found. Graftability is the availability of a semantically meaningful iterative core over which halting can be defined without inventing new architecture.

Family track	**# models**	**Representative iterative core (if any)**
Foundation / Adaptation	14	STATE (fixed transformer depth), AIDO.Cell (diffusion/flow heads), PertAdapt (GO-GNN loop)
Chemical / Drug	9	XPert (stacked transformer blocks); most single-shot autoencoders
KnowledgeGraph / GRN	8	TxPert, GEARS, PDGrapher (graph message-passing depth)
OT / Flow / Diffusion	7	scDiff (DDPM reverse chain), CellFlow (ODE solver steps), AlphaCell
Transfer / Spatial	5	SequenTx (episodic RL loop); mostly single-shot

Design decision. The catalog analysis pointed to a generative backbone (scDiff [10], then CellFlow [12]) as the theoretically cleanest primary host, with a graph backbone (TxPert) as a mechanistically independent cross-check. As we report next, that plan met an identifiability wall on every model we *trained ourselves*, and was ultimately resolved only by moving to a *frozen, externally pretrained* foundation model.

Why a bidirectional stack still satisfies the refinement criterion. STATE is a position-free bidirectional transformer with distinct weights per layer — on the face of it the *least* clean host on the "layer 3 is not a better version of layer 2" criterion that disqualified feed-forward decoders above. What licenses reading its layers as refinement rounds is a *residual-stream* property, not a shared-weight recurrence. STATE is a pre-norm residual network, $x_{k+1} = x_k + f_k(x_k)$: every layer writes an additive correction into the same residual stream that the frozen response readout consumes, so applying that one readout at any layer yields a valid endpoint estimate — the unrolled-iterative-estimation view of residual nets [3, 4]. A generic decoder fails the criterion for exactly the complementary reason: its intermediate activations live in an opaque feature space with no output-space readout, so "run it more" is undefined. The distinction we keep explicit is that STATE supplies refinement in this *iterative-inference* sense but *not* a shared-weight recurrent core; consequently our depth is a layerwise refinement-convergence estimate, not adaptive compute-time in the Graves/PonderNet sense (Section 4.2). We do not assert monotone layerwise refinement — and, empirically, it is not monotone (Figure 2); the method requires only that a per-perturbation convergence layer exists and be recoverable, which it is.

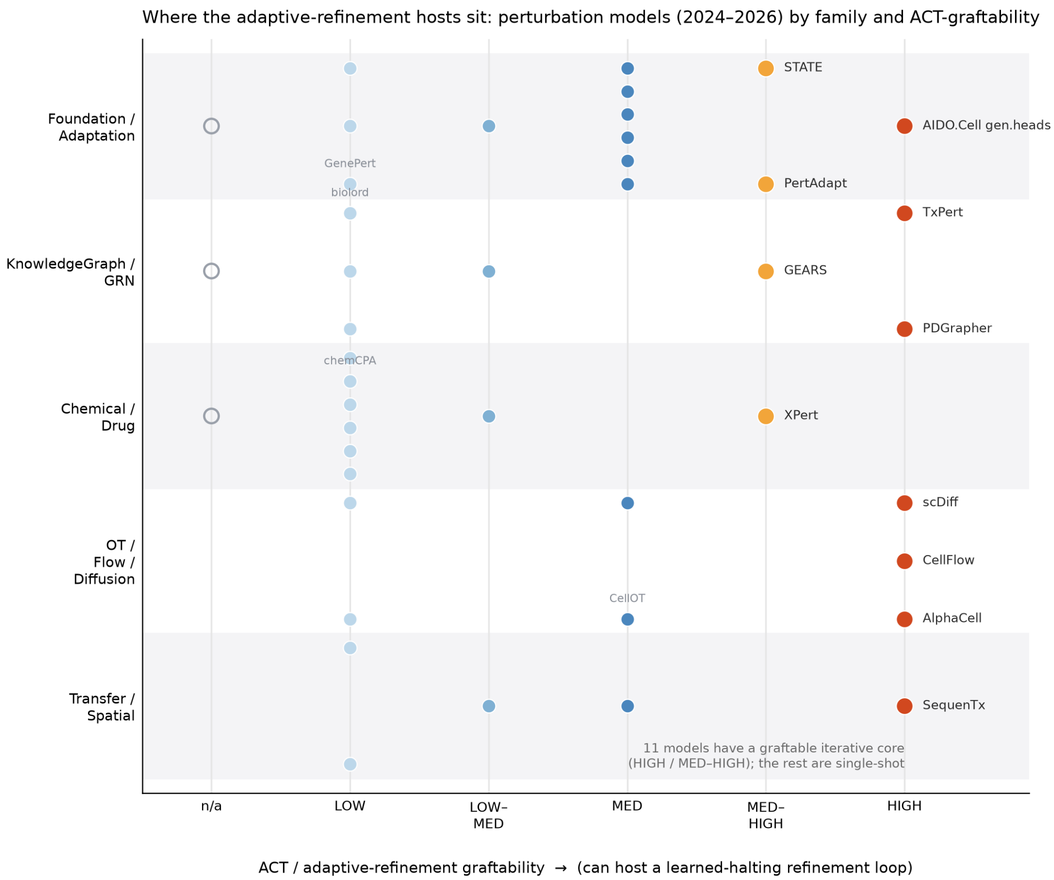

Figure 1: Where adaptive-refinement hosts sit. Perturbation models (2024–2026) by family track (rows) and ACT-graftability (columns). Highlighted models (STATE, scDiff, TxPert, CellFlow, GEARS, PDGrapher, AIDO.Cell heads) expose a semantically meaningful iterative core; the majority are single-shot and would require architectural changes to host a halting loop. Eleven of 43 models are graftable (HIGH / MEDIUM–HIGH).

3 The identifiability wall: negatives that shaped the method

Before the working method, we report the sequence of failures that produced it, because the failures are informative and they directly motivate each design choice in Section 4. We grafted learned halting onto several hosts — diffusion [10], graph message-passing [11, 13], and refiners we built ourselves — and, in every case, the learned depth landed in one of two failure modes. Table 2 summarises the outcomes against the two-corner test (reproducible *and* effect-independent).

Three rulings. Distilling the failures yields three rulings that constrain any method that hopes to clear both gates.

1. **A free learned gate over an expressive predictor collapses to constant depth.** A PonderNet halt head over pooled cell features, given every advantage (convergence-feature inputs, small-λ init, zero KL prior, entropy bonus, tens of epochs, multiple seeds), learns a *single global constant per seed.* On single-endpoint data the aggregate per-round loss is nearly flat, so the gradient toward any particular stopping round is weak, and the gate settles on whichever

Table 2: The identifiability wall. Every learned-halting graft onto a model we controlled either collapsed to a constant, degenerated into an effect-size proxy, or failed to reproduce. Only a frozen, externally pretrained foundation model reached the useful corner (reproducible *and* effect-independent). Values are Spearman ρ unless noted; "effect-independent" reports either ρ(depth,effect) or covariate R^2.

Host (data)	**Reproducible?**	**Effect-independent?**	**Verdict**
scDiff + PonderNet (Adamson)	depth $\approx$ constant	$\rho = -0.70$	effect-size proxy
TxPert adaptive-hops (Adamson)	$\rho = 0.71$	$\rho = -0.84$	effect-size proxy
Magnitude-free own refiner	$\rho = 0.14$	$R^2 = 0.09$	non-redundant but **irre**
Fixed-rule STRING cascade	$\rho = 0.96$	$R^2 = 0.04$	reproducible but a grapl
Biology-fused ACT on TxPert	$\rho = 0.71$	$\rho = -0.84$	effect-size proxy again
Frozen STATE (this work)	**0.84**	$R^2 = 0.07$**; resid-repro 0.94**	**useful corner**

constant its seed drifts to. There is no per-perturbation signal to key on unless the gate is given one.

2. **Anchoring the halt loss to response magnitude makes depth reproducible but redundant.** When the halting target is tied (directly or through a magnitude-sensitive loss) to the size of the response, depth becomes stable across seeds — but its correlation with effect size is strongly negative ($\rho \approx -0.7$ to -0.84), i.e. it is an effect-size proxy and adds nothing.

3. **Removing magnitude makes depth non-redundant but irreproducible.** A magnitude-free target (matching only the *direction* of the response) removes the effect-size proxy, but a model we trained ourselves then produces a depth that does not replicate across seeds ($\rho = 0.14$).

These rulings are the crux of the identifiability problem: on a single endpoint, the two desiderata pull against each other for any model whose depth is a transient of *our* training. The resolution has two ingredients. First, use a *magnitude-free per-round target* (cosine distance between predicted and true response directions) so depth cannot be an effect-size proxy by construction. Second, provide the halting head with a per-perturbation supervision signal — an *oracle stopping round* derived from the model's own per-round convergence — so a magnitude-free depth can nonetheless be learned reproducibly. Crucially, both ingredients are viable only when the underlying predictor is a *well-fitted, frozen* model whose layerwise representation of each perturbation is shaped by data far beyond a single endpoint — which is why the wall fell only on a pretrained foundation model, not on any model we trained on one endpoint.

4 Methods

4.1 Host model and data

The host is ARC Institute's pretrained **STATE Transition** perturbation foundation model [5] (`arcinstitute/ST-SE-Replogle`, K562 fewshot checkpoint), a real `StateTransitionPerturbationModel` whose weights we load from the ARC checkpoint with zero missing and zero unexpected tensors. The backbone is an 8-layer bidirectional (NoRoPE, position-free) Llama-style transformer with hidden size 328, operating in the SE-600M cell-embedding space (`X_state`, 2058-d); a gene decoder maps to 2000 highly-variable genes and predicts expression. **All ARC weights are frozen**: nothing about the transformer is retrained, so the depth structure we measure is a property of the pretrained model, not of our fit.

We use ARC's *native* K562 evaluation set [6] (`adata_real.h5ad`): 134,751 cells, 968 perturbations, with `X_state` precomputed. On these data the pretrained model predicts well — DE-gene response correlation 0.879 (median 0.929) against ground truth — which is a precondition for the analysis: a halting signature on a model that cannot predict the data would measure noise. We do not use Adamson, because ARC's model does not transfer to it out of the box (response-direction cosine +0.06 to +0.17 even for strong perturbations); measuring refinement depth there would be uninterpretable. The cross-line experiments (Section 5.2) repeat the identical construction on the HepG2, Jurkat, and RPE1 fewshot checkpoints.

4.2 Eight transformer layers as eight refinement rounds

STATE has no runtime-varied loop — it is a fixed 8-layer stack traversed in one forward pass, and all 8 layers always run. We treat the layer axis as the refinement axis: a single forward pass with `output_hidden_states=True` exposes all 8 layers' hidden states, and at each layer k we apply *the same frozen* ARC readout — `project_out` then `gene_decoder`, identical parameters at every layer — to the layer-k hidden state to obtain a per-layer gene-expression prediction $\hat{Y}_k$. This is a logit-lens readout of the residual stream, not a set of learned per-layer probes: because one fixed map decodes every layer, $\hat{Y}_k$ and $\hat{Y}_{k+1}$ are comparable points in output space and "layer $k+1$ is a better estimate than layer k" is well-defined (Section 2). This gives 8 candidate predictions per perturbation from one pass, at the cost of a modest reconstruction gap: the per-layer decode reaches ≈ 0.67 per-cell correlation with ARC's full-inference decode (the residual is basal-pairing detail in ARC's pipeline).

What the layers actually do (and what $\mathbb{E}[N]$ therefore is). Reading the frozen decode off each layer, the mean direction-error D_k over 924 K562 perturbations *refines* across the early stack — from $D_1 = 0.909$ down to a minimum $D_6 = 0.881$ — and then the last two layers move the readout *away* from the response direction ($D_7 = 0.919$, $D_8 = 0.939$) as the backbone's final layers specialise for its own pretraining objective rather than our endpoint readout (Figure 2a). Refinement is thus real but *not* globally monotone, and the per-perturbation convergence layer is genuinely dispersed: the layer of closest approach spans all eight, with mass on layers 1–2 (fast-converging perturbations) as well as 6–8 (Figure 2b). Two consequences follow. First, this is why depth must be *read*, not assumed — a fixed "use the last layer" rule would be strictly worse than the first layer for many perturbations. Second, it fixes what $\mathbb{E}[N]$ means: because the backbone is frozen and all eight layers always execute, $\mathbb{E}[N]$ varies no compute and saves none: it is the expected layer at which this fixed stack's endpoint estimate converges for a perturbation — a layerwise refinement-convergence depth, *not* adaptive compute-time in the Graves/PonderNet sense. This is the sharper form of the thesis's standing caveat: depth is a computational proxy for response complexity, and specifically a convergence-layer proxy, never biological time or dynamic compute.

Magnitude-free per-round target. For each perturbation, let the true response direction be $u = \text{normalize}(Y - Y_{\text{basal}})$ and the round-k predicted direction $\hat{u}_k = \text{normalize}(\hat{Y}_k - Y_{\text{basal}})$. The per-round loss is the *cosine distance between directions*,

$$D_k \;=\; 1 - \big\langle \hat{u}_k,\; u \big\rangle, \tag{1}$$

averaged over cells. Because D_k depends only on the direction and not the magnitude of the response, depth built on D_k cannot be an effect-size proxy by construction — this directly answers ruling 2 of Section 3.

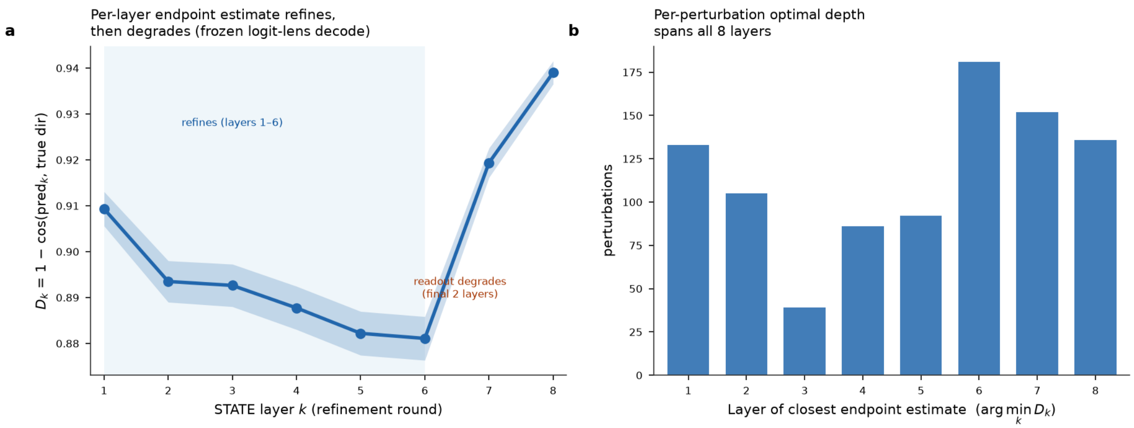

Figure 2: The frozen stack's per-layer endpoint estimate refines, then degrades — and the convergence layer is per-perturbation. (a) Mean per-layer direction error $D_k = 1 - \cos(\hat{u}_k, u)$ over 924 K562 perturbations (±SE): the frozen logit-lens decode refines through layers 1–6 (0.909→0.881), then the final two layers move the readout away from the response direction. Refinement is real but not monotone. (b) Per-perturbation layer of closest approach ($\arg\min_k D_k$) spans all eight layers, so $\mathbb{E}[N]$ draws on the full range rather than collapsing to one global round.

Computed readout. The simplest depth statistic is the per-layer loss argmin, $d = \arg\min_k D_k$ — which layer's prediction is closest to the true response direction. This is a *computed readout* of the frozen model, not a learned decision; we report it as an independent, parameter-free reference against which the learned depth is validated.

4.3 AdaptiveStateRefine: learned halting on the frozen backbone

The learned model, AdaptiveStateRefine, adds a halting head on top of the frozen backbone. Three design choices — each a direct response to the identifiability wall — make learned halting work where the naive gate collapsed.

1. **A refinement token (per-perturbation probe).** Instead of pooling cell features, a dedicated learnable token is appended to the cell sentence; because the backbone is position-free, the token appends cleanly. Its hidden state at each layer feeds the halt head, so the halting representation is built per-perturbation by the transformer itself rather than averaged from cell features. Diagnostically, this token's hidden state varies across perturbations at layers 1–6 (pairwise cosine 0.80–0.83), confirming a per-perturbation signal exists for the gate to key on — the missing ingredient behind ruling 1.
2. **A LayerNorm-unsaturated gate (load-bearing).** The frozen refinement-token hidden states have large norm; without normalisation, the pre-sigmoid halt logits have standard deviation $\sim$100+, the sigmoid saturates to exactly 0/1, hazards become binary, stop-mass becomes one-hot, and depth collapses to a constant. A `LayerNorm` on the halt-head input keeps logits in a learnable range. This single change turns the halt confidence from a pinned 1.000 into genuine soft distributions (mean 0.28).
3. **Oracle stopping-round supervision (Section 4.4).**

Sequential-hazard halting and mixture prediction. Let $\lambda_k = \sigma(\ell_k)$ be the halt hazard at round k from the halt head (ℓ_k the logit), with the last round forced to stop ($\lambda_R = 1$). The stop mass is the product of hazard and survival,

$$q_k \;=\; \lambda_k \prod_{j<k}(1-\lambda_j), \qquad \sum_{k=1}^{R} q_k = 1, \tag{2}$$

and the expected halting depth is $\mathbb{E}[N] = \sum_k q_k\, k$. The prediction is the *stop-mass-weighted mixture* of per-round decoded predictions, $\hat{Y}_{\text{mix}} = \sum_k q_k\, \text{sg}[\hat{Y}_k]$, where $\text{sg}[\cdot]$ is a stop-gradient: the per-round predictions are *detached* so that the only free parameters shaping the mixture are the stop masses (the halt head), not the frozen decoder. Consequently the task-loss gradient flows to the halt head and pushes stop mass onto the round that best predicts each perturbation. An auxiliary error head (softplus) predicts each round's own loss D_k. Halt confidence is defined as the concentration of the stop distribution, $1 - H(q)/\log R$, and is kept strictly separate from STATE's `ConfidenceToken` (inactive in this checkpoint).

4.4 Oracle stopping-round supervision

For each perturbation we define an oracle stopping round from the model's own per-round convergence:

$$r^{*} \;=\; \min\Big\{\, r : D_r \le D_{\min} + \tau\,(D_1 - D_{\min}) \,\Big\}, \qquad \tau = 0.05, \tag{3}$$

i.e. the first round that captures at least $(1-\tau) = 95\%$ of the attainable improvement over the first exit. r^{*} is the supervision target for the halt head; it is magnitude-free (derived from the direction-only losses D_r) and per-perturbation.

4.5 Joint calibration of the halt head

The halt head is trained with a jointly-calibrated loss combining a mixture task term, deep per-round supervision, oracle cross-entropy, a ponder cost, and an error-head term:

$$\mathcal{L} \;=\; D(\text{mix}) \;+\; \alpha\,\overline{D_r} \;+\; \beta\,\big(-\log q_{r^{*}}\big) \;+\; \gamma\,\frac{\mathbb{E}[N]}{R} \;+\; \delta\,\text{Huber}\big(\hat{e}_r,\, \text{sg}[D_r]\big), \tag{4}$$

with $\alpha = 0.5$, $\beta = 1.0$, $\gamma = 0.1$ (ponder applied only after a 15-epoch warm-up), $\delta = 0.1$, and the PonderNet KL prior set to zero. The β term pulls stop mass onto the oracle round r^{*}; the γ ponder term encourages earlier halting once the head has learned structure. All hyperparameters were fixed on held-out validation perturbations; the test set was never touched during selection.

Anti-circularity guardrail. There is deliberately *no* term coupling guides or perturbations that target the same gene to a shared depth. Such a term would make the depth signature reproducible by construction and would render any guide- or gene-level reproducibility test circular. We leave same-gene coupling to a future regulariser so that the reproducibility results below are earned, not imposed.

4.6 Evaluation: the three gates

We evaluate against the test order of Section 2. **(i) Reproducibility** is measured as split-half rank correlation: the halt head is trained independently on two disjoint halves of the cells (or across independent training seeds / control draws), and we report the Spearman ρ between the two resulting $\mathbb{E}[N]$ rankings. **(ii) Non-redundancy** is measured by the partial Spearman correlation

$\rho(\mathbb{E}[N], \text{effect} \mid n_{\text{DE}}, n_{\text{cells}})$ and by the fraction of $\mathbb{E}[N]$ variance explained by regressing on effect size, #DE genes, and cell count (R^2). **(iii) Interpretation** (Section 5.4) tests $\mathbb{E}[N]$ against biological covariates including knockdown strength and target baseline expression, using independent annotations. Effect size is the L2 norm of the response; #DE and cell counts are computed from the native eval data. All correlations are Spearman unless stated.

5 Results

5.1 K562: a reproducible, effect-independent, learned depth

On the 968 K562 perturbations, both the computed readout and the learned head clear the two gates, and a naive learned gate does not — the three outcomes together are the headline of the method.

The computed readout. The per-layer argmin depth spans the full range (layers 1–8, mean 4.9, coefficient of variation 0.45) and is reproducible across 8 independent basal draws at $\rho = 0.836$. It is weakly (not strongly) coupled to effect size ($\rho = -0.29$), covariates explain only 7% of it ($R^2 = 0.073$), and the partial correlation controlling #DE and cell count is -0.027 — effect-independent. Most tellingly, after regressing out effect size, #DE, and cell count, the *residual* depth — the part that is not an effect-size proxy — still reproduces across independent draws at $\rho = 0.94$. The signature carries real, reproducible information beyond response magnitude.

The naive learned gate collapses. A PonderNet halt head over pooled features, given every advantage (Section 3, ruling 1), learns a single global constant per seed: $\mathbb{E}[N] \in \{1, 8, 2, 1\}$ across four seeds with within-seed standard deviation 0.000. It finds no per-perturbation structure. This is the negative that the refinement token, LayerNorm gate, and oracle supervision are designed to overcome.

The oracle-supervised learned head works. With the full scheme (Section 4, 4 seeds $\times$ 50 epochs, held-out validation), the learned $\mathbb{E}[N]$ spreads genuinely (mean 4.58, std 1.18, range $[1.36, 6.96]$, i.e. 70% of the 8-round budget) and clears both gates (Table 3, Figure 3): reproducibility $\rho = 0.761$ (validation 0.742); it recovers the oracle target ($\rho(\mathbb{E}[N], r^*) = 0.607$) and the independent computed argmin ($\rho = 0.589$); and it is effect-independent (partial $\rho(\text{effect} \mid n_{\text{DE}}, n_{\text{cells}}) = -0.044$, covariate $R^2 = 0.153$). The raw correlation with effect size is -0.35, so effect size, #DE, and cell count together absorb nearly all of it — we report the partial, not the raw, and note that the learned head carries slightly more effect leakage than the cleaner computed readout, as expected since r^* is derived from loss curves that depend weakly on effect size. Halt confidence averages 0.277 (soft distributions, not one-hot). This is a genuine learned adaptive-depth signature, obtained only with the oracle scaffold; a free gate does not discover it.

Illustrative biology. Early-exit (shallow) perturbations are dominated by ribosomal- protein and core-translation knockdowns (RPS26, RPL23, RPL31, RPL14) that the model captures immediately; late-exit (deep) perturbations include rRNA-processing, nuclear-import, and chromosome-maintenance factors (GAR1, NIP7, RNPC3, CD3EAP, SMC2, IPO7, MRTO4) that engage the full transformer depth. Depth is not rank-order effect size ($\rho = -0.29$). We keep the reading strictly computational: depth is refinement complexity in the pretrained model's representation, not biological time or causal cascade length.

Table 3: K562 ($n = 968$): the oracle-supervised learned $\mathbb{E}[N]$ clears both gates, alongside the computed argmin readout for reference. Both are reproducible and effect-independent; the naive learned gate collapses to a per-seed constant.

Property	Computed argmin	Learned $\mathbb{E}[N]$ (oracle)	Naive gate
Reproducibility ρ (split-half)	0.836	0.761 (val 0.742)	collapse
Depth spread	1–8, mean 4.9	mean 4.58, std 1.18	constant per seed
Recovers oracle r^*	—	0.607	0
Recovers computed argmin	—	0.589	0
Raw $\rho(\mathbb{E}[N], \text{effect})$	−0.29	−0.35	—
Partial $\rho(\text{effect} \mid n_{\text{DE}}, n_{\text{cells}})$	−0.027	−0.044	—
Covariate R^2	0.073	0.153	1.0 (trivial)
Residual reproducibility ρ	0.94	—	—

5.2 Cross-line generalisation: reproducible within, not across, cell types

We repeated the identical construction on all four Replogle lines (HepG2, Jurkat, K562, RPE1), each with its own frozen ARC checkpoint, and asked whether the depth ranking is a portable property of a perturbation or specific to a cellular context.

Within-line reproducibility is strong. Head-seed reproducibility is clear in every line (ρ= 0.85 HepG2, 0.76 Jurkat, 0.76 K562, 0.83 RPE1). The strongest and most honest reproducibility measure is the *learned*-$\mathbb{E}[N]$ *split-half*, in which the halt head is trained separately on each cell-half's response and we compare the model's $\mathbb{E}[N]$ across halves. This averages ρ= 0.80 (HepG2 0.87, Jurkat 0.73, K562 0.76, RPE1 0.84; Figure 4), and it sits *below* the effect-size split-half noise ceiling (mean ρ= 0.97; per line 0.94–0.98) — so $\mathbb{E}[N]$ is reproducible but not merely a noiseless re-encoding of effect size, and it far exceeds the oracle-depth (data-derived) split-half of ρ= 0.51.

The ranking does not port across cell types. On the 194 perturbations shared by all four lines, the cross-line $\mathbb{E}[N]$ agreement averages only ρ= 0.14 (pairwise range −0.17 to +0.51; Figure 5). This is not an artifact of the learned head: the model-free computed depth has an even lower cross-line ceiling (ρ= 0.022 for argmin, 0.011 for the oracle r^*). The depth a perturbation "needs" is therefore a property of the perturbation *in a given cellular context*, not a context-invariant property of the perturbation itself.

5.3 Donor stability: a decisive negative for portability

The cross-line result leaves open whether depth is at least stable across biological replicates of the *same* context. We tested this on primary human $CD4^+$ T cells (GWCD4i CRISPRi Perturb-seq), fitting a pooled three-context model (Rest, Stim-8hr, Stim-48hr; 34,033 gene×context units, 11,526 genes) and performing leave-one-donor-out across four donors: for each held-out donor we refit and read $\mathbb{E}[N]$, then correlated $\mathbb{E}[N]$ across folds.

The cross-fold donor stability averages ρ= 0.057, with individual donor pairs ranging from −0.331 to +0.524 (Figure 6b) — indistinguishable from zero on average. This completes a striking **three-tier decay** of reproducibility as the source of variation moves from the halt head to the underlying biology (Figure 6a):

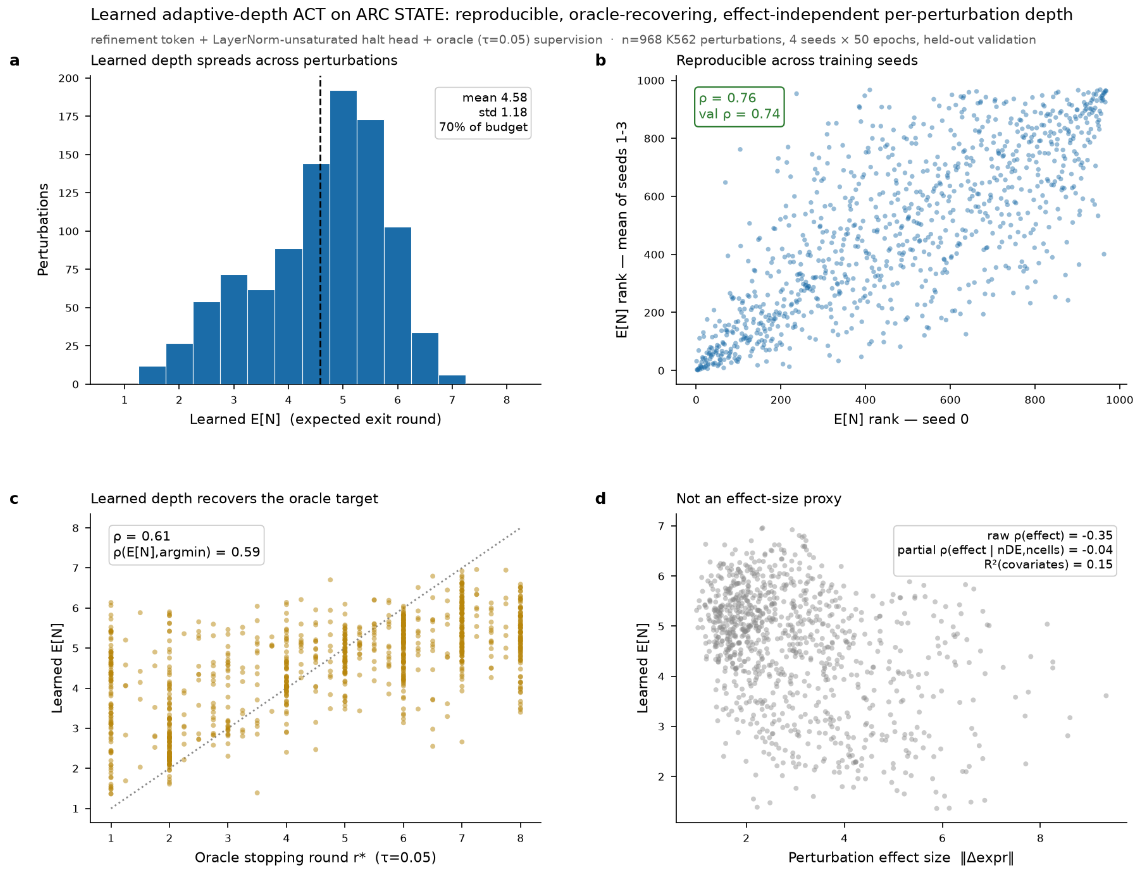

Figure 3: Oracle-supervised learned ACT on frozen STATE (K562, $n = 968$). (a) Learned $\mathbb{E}[N]$ spreads across perturbations (mean 4.58, 70% of the 8-round budget). (b) Reproducible across training seeds (ρ= 0.76; validation 0.74). (c) Recovers the oracle stopping round r^* (ρ= 0.61) and the computed argmin depth (ρ= 0.59). (d) Not an effect-size proxy: raw $\rho(\text{effect}) = -0.35$ but partial $\rho(\text{effect} \mid n_{\text{DE}}, n_{\text{cells}}) = -0.04$ (R^2 covariates $= 0.15$).

What varies between the two fits	$\mathbb{E}[N]$ **reproducibility** ρ
Only the halt-head seed (same backbone, same data)	0.93
The backbone seed (same data, retrained backbone)	0.29
The donor subset (different donors)	0.06

The signature is highly reproducible when only the halt head is reseeded, degrades when the backbone is retrained from scratch, and collapses across donors. The reading: on a fixed pretrained backbone — the way the model is actually deployed — the depth signature is *cell-type-specific*, reproducible within a context but not shared across contexts. The backbone-seed tier (0.29) shows a secondary dependence on the particular fit when the backbone itself is retrained from scratch, which is a robustness caveat rather than the deployment case. Portability across donors is a decisive negative, and it is the expected behaviour of a context-specific property, not evidence that depth is a mere training artifact.

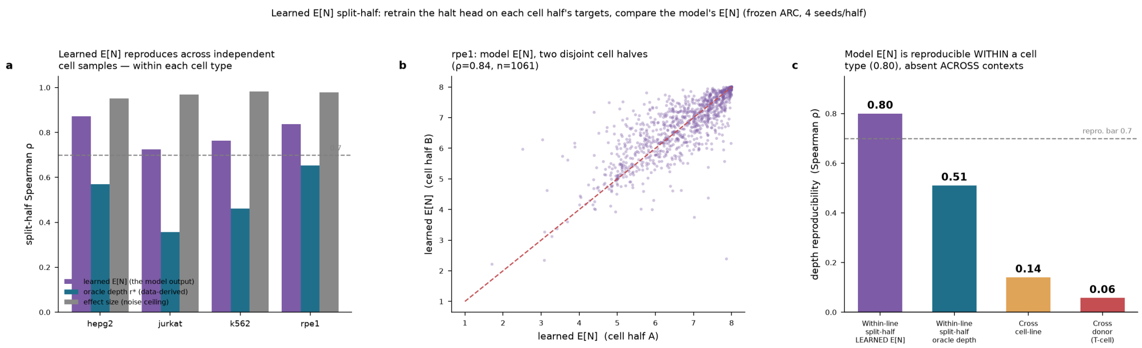

Figure 4: Learned-$\mathbb{E}[N]$ split-half reproducibility. The halt head is retrained on each cell-half's targets and the model's $\mathbb{E}[N]$ compared across halves (frozen ARC, 4 seeds/half). (a) Within each cell type the learned $\mathbb{E}[N]$ reproduces (purple), between the data-derived oracle depth (teal) and the effect-size noise ceiling (grey). (b) RPE1 example ($\rho = 0.84$, $n = 1061$). (c) The signature is reproducible *within* a cell type (mean 0.80) but decays across contexts: cross-line 0.14, cross-donor 0.06.

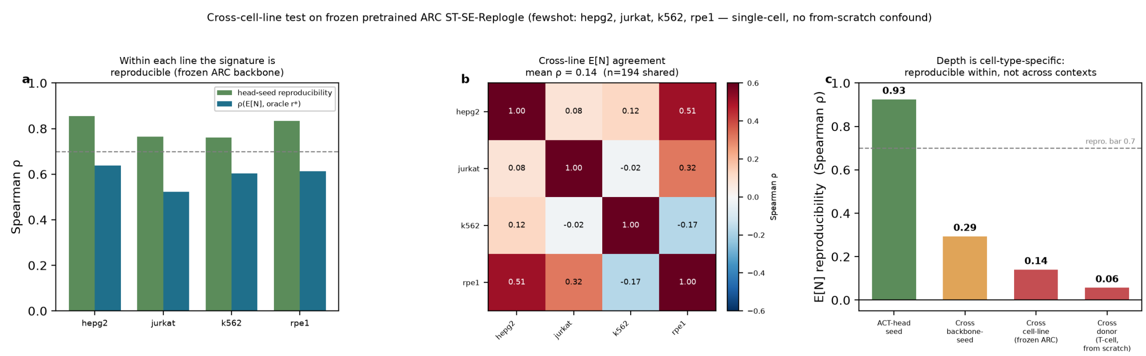

Figure 5: Cross-cell-line test on frozen pretrained ARC ST-SE-Replogle (fewshot: HepG2, Jurkat, K562, RPE1; single-cell, no from-scratch confound). (a) Within each line the signature is reproducible (head-seed $\rho = 0.76$–0.85). (b) Cross-line $\mathbb{E}[N]$ agreement is weak (mean $\rho = 0.14$, $n = 194$ shared perturbations). (c) Depth is cell-type-specific: reproducible within a line but not portable across contexts. Reproducibility decays from head-seed (0.93) through backbone-seed (0.29) and cross-cell-line (0.14) to cross-donor (0.06).

5.4 Biological validation across all four lines

To characterise what depth tracks biologically — and to confirm effect-independence beyond K562 — we correlated $\mathbb{E}[N]$ against five covariates in every line: response magnitude, #DE genes, cell count, knockdown strength (the autologous target's own $|\Delta|$), and target baseline expression, plus the partial correlation depth$\sim$effect controlling $\{n_{\mathrm{DE}}, n_{\mathrm{cells}}\}$ (Table 4, Figure 7).

Depth is not a reagent or abundance artifact. Knockdown strength and target baseline expression are uncorrelated with $\mathbb{E}[N]$ in *every* line ($|\rho| < 0.10$, all n.s.). Halting depth is therefore not measuring assay efficiency or gene abundance — it is a property of the *response*, not the *reagent*. This is the cleanest negative control in the panel and it passes in all four lines.

Depth relates to complexity with a cell-type-specific sign. $\mathbb{E}[N]$ correlates with response magnitude and #DE genes ($\rho = 0.24$–0.36) in HepG2, Jurkat, and RPE1 — larger, broader responses need more refinement — but K562 has the *opposite* sign (-0.35): there, larger responses

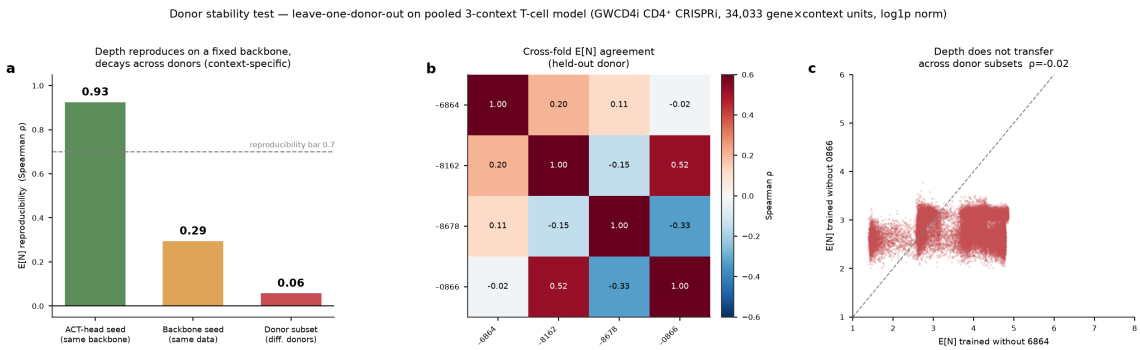

Figure 6: Donor stability — leave-one-donor-out on a pooled 3-context T-cell model (GWCD4i CD4$^+$ CRISPRi, 34,033 gene×context units). (a) $\mathbb{E}[N]$ reproduces within a fitted model (halt-head seed 0.93) but decays to zero across donors (0.06), through an intermediate backbone-seed tier (0.29). (b) Cross-fold $\mathbb{E}[N]$ agreement matrix across the four held-out donors (mean $\rho = 0.06$; individual pairs -0.33 to $+0.52$). (c) Depth does not transfer across donor subsets ($\rho = -0.02$ for the illustrated pair).

Table 4: Biological validation, all four lines. Spearman ρ of $\mathbb{E}[N]$ against each covariate (full panel $n \approx 945$–1084; knockdown strength and baseline expression require matching a perturbation to its own gene's row, so use the target-in-panel subset $n \approx 169$–223). $^{***}p < 10^{-3}$, $^{*}p < 0.05$; unmarked $p > 0.05$.

Covariate	K562	RPE1	Jurkat	HepG2
Response magnitude	-0.35^{***}	$+0.32^{***}$	$+0.24^{***}$	$+0.36^{***}$
# DE genes	-0.35^{***}	$+0.31^{***}$	$+0.23^{***}$	$+0.32^{***}$
Cell count	-0.08^{*}	-0.24^{***}	-0.01	$+0.14^{***}$
Knockdown strength	-0.03	$+0.10$	-0.00	$+0.02$
Baseline expression	-0.03	$+0.09$	-0.00	$+0.03$
Partial (depth∼effect | nDE, nCells)	$+0.03$	-0.02	$+0.07$	$+0.16^{*}$

halt earlier. This sign flip is itself a cell-type-specificity result: the depth↔complexity relationship is not universal, consistent with the cross-line $\rho = 0.14$ of Section 5.2.

Effect-independence holds in all four lines. After regressing out #DE and cell count, the partial correlation between depth and effect size collapses to $+0.03$ (K562), -0.02 (RPE1), $+0.07$ (Jurkat), and $+0.16$ (HepG2, the only weak residual, $p = 0.03$). The raw depth↔magnitude correlation is thus mediated by DE-gene count, not by a direct dependence on effect size — the K562 anchor (partial -0.04) reproduced across the panel.

5.5 Halting-depth clusters organize druggability and toxicity

Sections 5.1–5.3 treat $\mathbb{E}[N]$ as a scalar and test it *linearly* against covariates. That view is deliberately conservative, and it is also where a real pattern hides. When perturbations are **clustered on the depth signature itself** — the four-seed $\mathbb{E}[N]$ fingerprint with halt confidence and the oracle round r^*, reduced by UMAP and partitioned by Leiden into 9–11 clusters per line — and each cluster is cross-referenced against the 1708-gene drug/toxicity annotation, the clusters resolve into functional strata with sharply different pharmacological profiles.

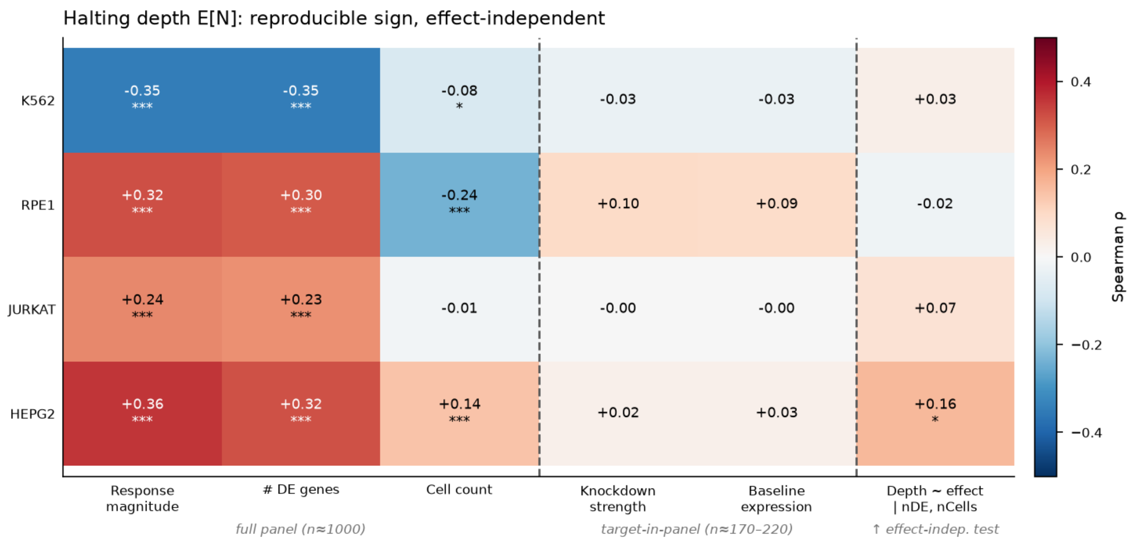

Figure 7: Halting depth $\mathbb{E}[N]$: reproducible sign, effect-independent (all four lines). Left block (full panel): depth co-varies with response magnitude and #DE genes, with a *cell-type-specific sign* (K562 negative; RPE1/Jurkat/HepG2 positive), and only weakly with cell count. Middle block (target-in-panel): depth is uncorrelated with knockdown strength and baseline expression in every line. Right column: the effect-independence test — partial correlation depth∼effect controlling #DE and cell count — collapses toward zero in all lines (weak residual only in HepG2).

The depth axis recovers functional gene classes. Labelling each Leiden cluster by its most over-represented GO-Biological-Process theme ($\geq 1.3\times$ enrichment vs the line background) gives coherent, data-driven identities: the deepest cluster is **translation/ribosome-dominated** in HepG2 (4.1×), Jurkat (4.1×), and RPE1 (4.5×); DNA-replication/repair and mRNA-splicing occupy mid-depth clusters; transcription-regulation sits shallowest (HepG2 3.7×). K562 is inverted, as everywhere else in this paper: its translation/ribosome cluster is the *shallowest* ($\mathbb{E}[N] = 2.0$, 2.2×), mirroring the depth sign flip of Section 5.4.

Approved and clinical drug targets concentrate in that translation/ribosome cluster — in every line. This is the strongest single pharmacological pattern in the dataset (Table 5), and it is invisible to the linear test: the correlation between $\mathbb{E}[N]$ and approved/clinical status is weak in every line ($|\rho| \leq 0.21$), because the drug-target density is not spread monotonically along the depth axis — it is concentrated in one functional cluster sitting at whichever depth extreme the ribosomal machinery occupies. Per-cluster, the fraction of approved/clinical drug targets peaks in the translation/ribosome cluster at odds ratios of 2.8–9.8 over the rest of the line, with HepG2 reaching 38% approved/clinical (Fisher $p = 6 \times 10^{-14}$).

Toxicity and essentiality co-localize in the same clusters. These drug-dense clusters are also the most safety-flagged and most essential: in HepG2 the deep translation cluster is 97% common-essential (OR 13.6) and 90% safety-flagged (OR 5.8); Jurkat's c9 translation cluster is 97% essential (OR 12.9); K562's shallow translation cluster is 94% safety-flagged (OR 9.4); RPE1's deep c6 is 83% safety-flagged (OR 3.2). The depth signature therefore separates perturbations into strata that differ jointly in druggability *and* toxicity liability, and the drug-and-toxicity-dense stratum sits at whichever depth extreme the translation/ribosome machinery occupies in that line.

Figure 8: Depth-signature UMAP with Leiden clusters annotated by fold-enriched GO-BP theme and mean halting depth $\mathbb{E}[N]$, per Replogle line. The $\mathbb{E}[N]$ gradient runs smoothly across each manifold; translation/ribosome clusters anchor the depth extreme — deepest in HepG2/Jurkat/RPE1, shallowest in K562.

Table 5: Approved/clinical drug targets concentrate in the translation/ribosome depth cluster (Fisher exact vs the rest of the line).

Line	Cluster (depth position)	mean $\mathbb{E}[N]$	% approved/clinical	Odds ratio	p
HepG2	c8 (deepest)	8.0	38%	9.8	6×10^{-14}
RPE1	c6 (deepest)	8.0	31%	5.7	1×10^{-8}
K562	c8 (shallowest)	2.0	29%	5.6	6×10^{-6}
Jurkat	c8 (deep)	7.7	18%	2.8	7×10^{-3}

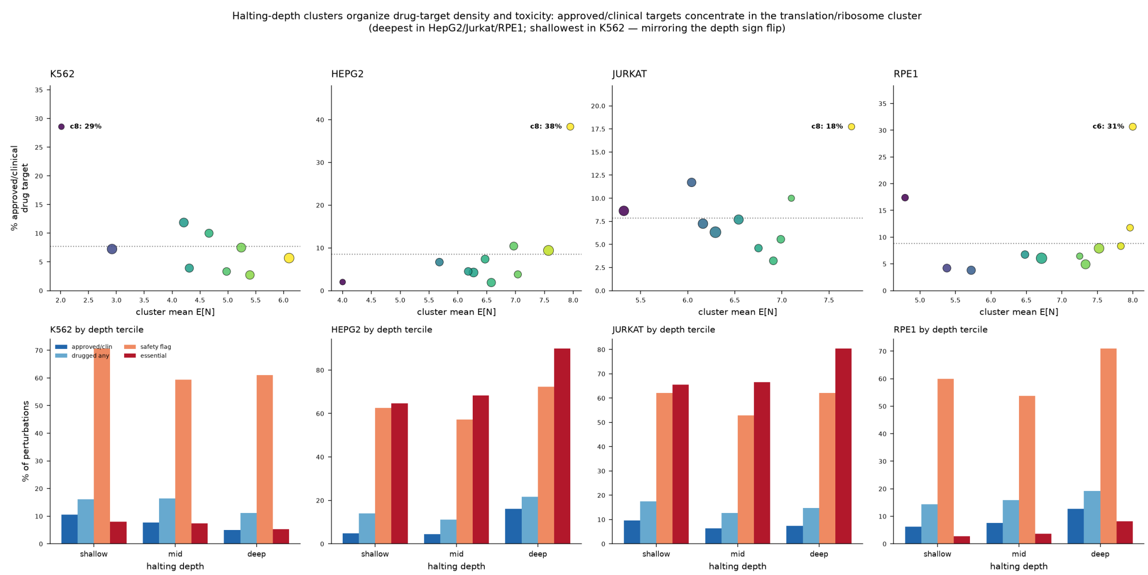

Figure 9: Halting-depth clusters organize drug-target density and toxicity. Top row: per cluster, the fraction of approved/clinical drug targets peaks in the translation/ribosome cluster at the depth extreme (dotted line = line background). Bottom row: by depth tercile, essentiality and safety-flag density rise toward the deep end in HepG2/Jurkat/RPE1 and toward the shallow end in K562, tracking where the ribosomal machinery sits.

Interpretation. The organization is real and reproducible across all four lines, and it is mechanistically coherent: the translation/ribosome cluster is simultaneously drug-target-rich (many established oncology targets act on translation and ribosome biogenesis), essential, and safety-flagged, and the halting depth places it at a reproducible extreme of the refinement axis. This is a genuine structuring of the target landscape by the depth signature — not merely a restatement of effect size, since the linear effect-size and depth correlations are weak (Section 5.1). It is descriptive rather than causal: depth clusters provide a *map of where druggable-but-toxic machinery sits on the refinement axis*, useful for stratifying candidates by both opportunity and liability. Critically, this signal is recovered from the *depth signature alone*; Section 5.6 shows that once a STRING-network baseline is added, the same biology is absorbed by the network prior and $\mathbb{E}[N]$ shows no incremental value — the network encodes the identical module structure, so the two are redundant, not contradictory.

Table 6: ACT value-add for drug-target discovery, per line. Three of four lines are negative; RPE1 is a weak, drug-class-specific positive. "redundant" = $\mathbb{E}[N]$ separates approved-vs-undrugged in-sample but adds nothing out-of-sample; "irrelevant" = $\mathbb{E}[N]$ does not separate them at all.

Line	Verdict	ΔAUC from $\lvert\Delta\mathbb{E}[N]\rvert$	# candidates	# anchors
HepG2	redundant	-0.010 (gene-disjoint CV)	798	70
Jurkat	irrelevant	-0.002 ($p = 0.77$); STRING $+0.024$	922	64
K562	redundant	≈ 0 out-of-sample	572	55
RPE1	adds (weak)	$+0.027$ (CI excl. 0; LR $p = 1.9\times10^{-7}$)	886	76

5.6 Incremental value of depth over a response-plus-network prior

Having characterised $\mathbb{E}[N]$ as reproducible, effect-independent, and cell-type-specific, we asked whether it is *useful* for a concrete downstream task: prioritising undrugged perturbations as candidate members of an approved or clinical drug-target class. For each line we built a discovery pipeline that clusters perturbations by fused response and STRING-network [14] similarity, then ranks undrugged perturbations by their similarity and STRING adjacency to approved/clinical *anchor* targets. The tested hypothesis is precise: *does adding the ACT halting axis $\lvert\Delta\mathbb{E}[N]\rvert$ improve drug-target-class discrimination beyond response and network similarity?* Annotation over the 1708-gene union (74 fields) is Claude-generated from public databases (Open Targets [15], ChEMBL, gnomAD, DepMap, Pharos, STRING, FDA) with per-value source tags — it is not hand-curated, and every candidate is a hypothesis ("same class as anchor X"), not a validated target.

The value-add verdict: three of four lines negative. Adding $\lvert\Delta\mathbb{E}[N]\rvert$ does not improve target-class identification in HepG2 (ΔAUC $= -0.010$, gene-disjoint CV), Jurkat (ΔAUC $= -0.002$, $p = 0.77$; whereas STRING adds $+0.024$, $p = 0.001$), or K562 (ΔAUC ≈ 0 out-of-sample beyond response and effect). RPE1 is a weak, drug-class-specific positive (ΔAUC $= +0.027$, bootstrap CI excludes 0, likelihood-ratio $p = 1.9 \times 10^{-7}$), where $\mathbb{E}[N]$ acts as a tie-breaker among DNA-replication-machinery candidates rather than a primary driver (Table 6). The verdict held identically across two annotation versions (before and after merging richer per-line essentiality and additional drug anchors), so it is not an artifact of anchor choice. Importantly, $\mathbb{E}[N]$ is *not* merely redundant with response magnitude ($\lvert\rho(\mathbb{E}[N], \text{effect})\rvert$ is low by design); it simply carries little *target-class* information — response complexity is largely orthogonal to pharmacological target class.

This linear negative should be read alongside the positive result of Section 5.5, not against it. The value-add test here adds $\mathbb{E}[N]$ *on top of a STRING-network baseline*, and STRING is a curated protein-interaction graph that already encodes the functional-module structure distinguishing the target classes — indeed STRING itself contributes the only significant lift (ΔAUC $+0.024$, $p = 0.001$ in Jurkat). Because depth clustering (Section 5.5) recovers those same modules directly — the translation/ribosome stratum, the replication stratum — the depth signal and the network prior are *redundant*: once the network is in the baseline, $\mathbb{E}[N]$ has no orthogonal target-class information left to add. The correct reading is that the depth signature *alone* organizes the druggability/toxicity landscape (Section 5.5); it does not do so *incrementally over a network prior built from the same underlying biology.*

What the ranking does deliver. Independent of the ACT axis, the response+STRING ranking produces coherent, mechanism-consistent hypotheses: in HepG2, RPA2/RPA1 $\rightarrow$ ATR (ssDNA-binding replication stress) and BORA $\rightarrow$ PLK1; in Jurkat, TTI1/SEH1L $\rightarrow$ MTOR and UHRF1 $\rightarrow$ DNMT1; in K562 and RPE1, obligate DNA-replication and OXPHOS complex partners anchored on approved targets such as PRIM1. The pipeline's value lies in the ranking, not the halting axis.

Cross-line consistency is confounded by essentiality. Across 1415 unique candidates, 100 recur in all four lines, 439 in three, 585 in two, and 291 in one — but this recurrence is dominated by essentiality: 1026 of 1415 candidates are core-essential in every nominating line. These recur precisely because core-essential ribosome-biogenesis, DNA-replication, and OXPHOS genes produce large, similar responses in every line, and they are exactly the genes one would *not* inhibit therapeutically. The defensible recurrent hypotheses are the 378 safety-pass candidates (non-essential in all nominating lines). The standout is MIOS $\rightarrow$ MTOR (all four lines, mean score-percentile 0.90, non-essential; a GATOR2-complex regulator of mTORC1), corroborated by LAMTOR1 $\rightarrow$ MTOR (three lines; Ragulator complex). These mTOR-regulatory hits are the clearest cross-line-robust, non-toxic, mechanistically-coherent signal — and, consistent with the non-portability of $\mathbb{E}[N]$, they emerge from the response+network ranking, not from the halting axis.

5.7 Non-redundancy with network topology: depth is not a recomputed graph statistic

A natural skeptic's objection is that halting depth $\mathbb{E}[N]$ — which we interpret as a proxy for response complexity, how far a perturbation's effects propagate through downstream regulatory cascades — might be an expensive way to recompute a quantity that is free to read off a graph. If a model-free network statistic (cascade depth, downstream reach, hub out-degree) reproduced the per-perturbation ordering of $\mathbb{E}[N]$, then the learned depth would be redundant with wiring the field already has. We test this directly.

Baselines. We build two networks. The *directed* gene-regulatory network is CollecTRI (via OmniPath): 45,553 TF$\rightarrow$target edges, 1,176 TFs. From it we compute, for each perturbed gene, out-degree (hub-ness), BFS cascade depth (longest downstream shortest-path), downstream reach (descendant count), reach-within-2, in-degree, betweenness, and PageRank. The *undirected* baseline is STRING v12 high-confidence (combined score $\geq$ 700, 234,370 edges), giving PPI degree, mean shortest path, and clustering. We then correlate each statistic with $\mathbb{E}[N]$ per cell line (Spearman ρ), and compute the partial correlation controlling for effect size and DE breadth to separate topology from raw response magnitude.

No network statistic reproduces $\mathbb{E}[N]$. The theory-motivated directed statistics are essentially uncorrelated with depth. Cascade depth gives $\rho = -0.01$ (K562), -0.14 (HepG2), -0.04 (Jurkat), -0.01 (RPE1); out-degree is likewise near zero in every line ($|\rho| \leq 0.13$). Across all ten statistics and four lines the single strongest correlate is PPI degree at $|\rho| = 0.23$ (K562) — and even its sign is inconsistent across lines (K562 -0.23 but HepG2 $+0.19$), the opposite of what a genuine shared determinant would produce. Every line's best network statistic sits *below* a $|\rho| = 0.3$ novelty ceiling and nowhere near a $|\rho| = 0.5$ redundancy floor (Figure 10). Partial correlations controlling for effect size and n_{DE} leave these near zero (max $|\rho_{\mathrm{partial}}| = 0.14$), so $\mathbb{E}[N]$ is not a re-encoding of response magnitude either. Halting depth carries per-perturbation ordering that neither network topology nor effect size predicts — it is a genuinely novel descriptor, the non-redundant half of the "useful" claim.

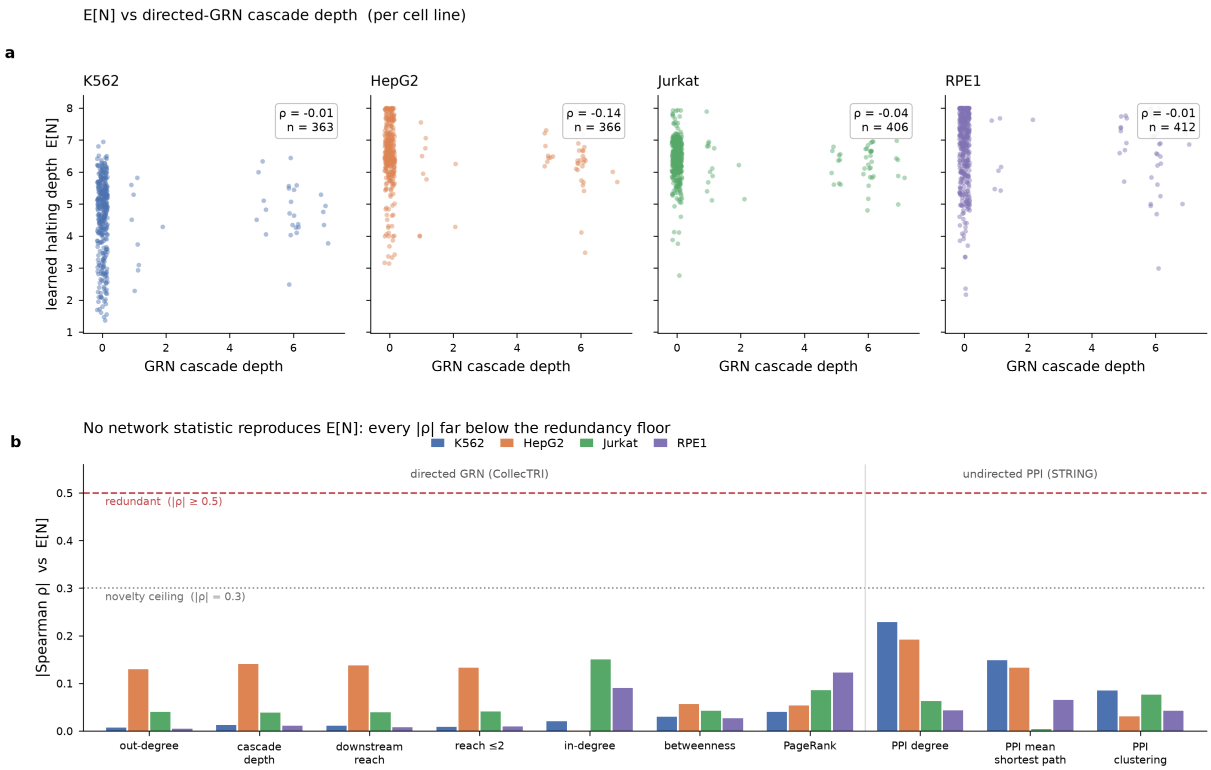

Figure 10: Halting depth $\mathbb{E}[N]$ is non-redundant with network topology. (a) $\mathbb{E}[N]$ versus directed-GRN cascade depth, per line: no relationship (ρ from -0.01 to -0.14). (b) Absolute Spearman ρ between $\mathbb{E}[N]$ and each of ten network statistics (directed CollecTRI GRN, left; undirected STRING PPI, right), all four lines. Every bar falls below the $|\rho| = 0.3$ novelty ceiling; none approaches the $|\rho| = 0.5$ redundancy floor. The strongest correlate is PPI degree at $|\rho| = 0.23$ (K562), with inconsistent sign across lines.

Caveat: directed coverage. CollecTRI is TF$\rightarrow$target only, so the directed downstream statistics are identically zero for the $\sim$88% of in-network perturbations that appear solely as targets; the directed test is under-powered and tie-dominated. Restricting to the 78 perturbations that are TFs with ≥ 1 downstream target, $\mathbb{E}[N]$ versus out-degree gives $\rho = 0.38$ (K562, $p = 0.02$), 0.46 (HepG2, $p = 0.001$), -0.01 (Jurkat), 0.26 (RPE1) — a modest, *inconsistent* positive in two of four lines, still well below redundancy. The well-powered undirected PPI comparison (n$\approx$900–1030 per line) is the primary result and is unambiguous.

5.8 Application: CD4$^+$ T-cell inflammatory-program suppressors at single-cell resolution

The second thesis asks a more specific question in primary human CD4$^+$ T cells: do perturbations that suppress stimulated inflammatory programs while sparing resting cells share refinement signatures distinct from damaging, inflammatory, or unstable perturbations? We answer it on the Marson/Pritchard genome-wide CD4$^+$ CRISPRi Perturb-seq screen (GWCD4i), at *single-cell* resolution, across two donors (D1, D4) and three conditions (resting, 8 h and 48 h αCD3/αCD28 stimulation). For each condition we warm-start the frozen ST-HVG-Replogle backbone, train per-condition response adapters and the oracle-supervised halting head, and extract the per-perturbation signature ($\mathbb{E}[N]$, halt confidence, nonlinear correction, predicted error) for 6,500–7,900 perturbations, each with ≥ 30 cells (2000 HVG, cell-load normalization). This replaces our earlier pseudobulk attempt, whose substrate was donor-unstable by construction (raw CD4 pseudobulk response $\rho\approx 0.07$–0.09 across donors, Section 5.3); the single-cell design recovers the signal the pseudobulk could not.

The signature is reproducible within a donor, and reproducibility rises with stimulation. A random cell-half split within each condition — the halt head trained separately on each half's response targets — gives split-half $\mathbb{E}[N]$ agreement of $\rho = 0.62$ (resting), 0.68 (8 h), and 0.75 (48 h). At single-cell resolution the per-perturbation depth signature is thus highly reproducible within a donor, far above the pseudobulk donor floor ($\rho\approx 0.06$) and inside the Replogle within-line range (0.72–0.87). Reproducibility increases monotonically as the cells are driven toward a more constrained activation state.

Depth is donor-specific, except at the stimulated endpoint. The decisive portability test correlates per-perturbation $\mathbb{E}[N]$ across the two donors under the same frozen backbone. In the resting and early-activation states the signature does *not* transfer (cross-donor $\rho = -0.11$ resting, -0.10 at 8 h; $n \approx 5{,}700$–$6{,}100$ shared perturbations), but at the fully-stimulated 48 h endpoint it becomes substantially portable ($\rho = 0.49$, $n = 5{,}840$, $p \approx 0$). The reading is biological: resting and early-activation cell states are donor-idiosyncratic, so perturbation-response geometry — and hence refinement depth — is donor-specific; by 48 h of strong stimulation, cells from different donors converge onto a shared activation program, and in that common state response complexity becomes a reproducible, donor-transferable property. This mirrors the Replogle cross-line result (Section 5.2) and is fully consistent with the paper's central claim that depth is cell-type/context-specific rather than a portable per-perturbation invariant — the signature ports exactly where, and only where, the biology converges.

Resting-sparing suppressors carry a distinct, shallower refinement signature. We define a data-driven activation axis as the NTC contrast between resting and 48 h-stimulated controls; it

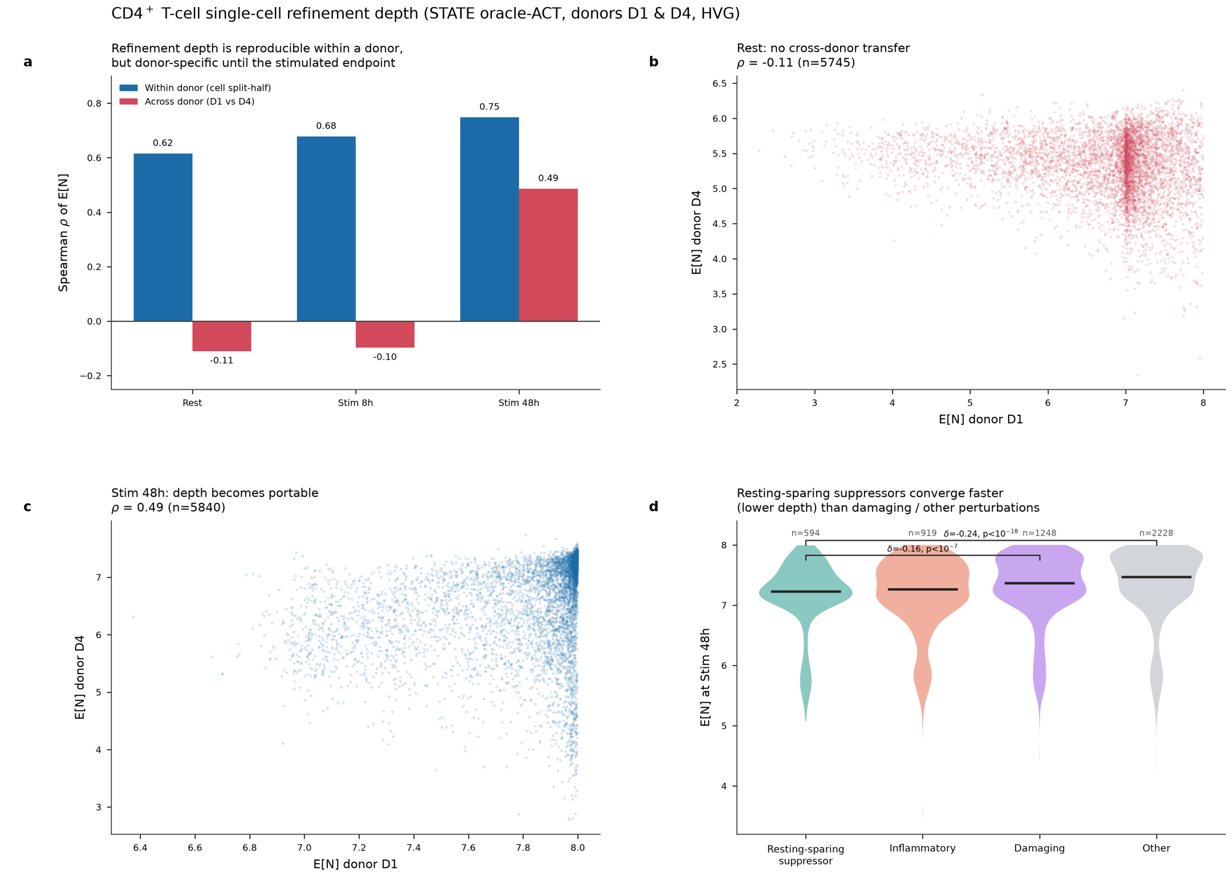

Figure 11: CD4$^+$ T-cell single-cell refinement depth. STATE oracle-ACT on GWCD4i CRISPRi Perturb-seq, two donors, three conditions, 2000 HVG. (a) Within-donor split-half $\mathbb{E}[N]$ reproducibility is high and rises with stimulation (0.62/0.68/0.75), while cross-donor agreement is ≈ 0 in resting and early activation but reaches $\rho = 0.49$ at the 48 h endpoint. (b,c) Cross-donor $\mathbb{E}[N]$ scatter: no transfer at rest ($\rho = -0.11$), substantial transfer at Stim 48h ($\rho = 0.49$). (d) Refinement depth separates phenotype categories at Stim 48h (Kruskal–Wallis $p = 5 \times 10^{-28}$): resting-sparing suppressors converge faster (lower $\mathbb{E}[N]$) than damaging ($\delta = -0.16$) and generic ($\delta = -0.24$) perturbations. n per group annotated.

loads on canonical T-cell activation genes (top-loading *IL2RA*, *DDIT4*, *CCL3*, *CCL4*, *LAG3*). Projecting each perturbation's response onto this axis and onto its resting-state engagement yields four categories: resting-sparing suppressors ($n = 594$), inflammatory ($n = 919$), damaging ($n = 1{,}248$), and other ($n = 2{,}228$). At the 48 h endpoint, $\mathbb{E}[N]$ separates these categories (Kruskal–Wallis $p = 5 \times 10^{-28}$). Resting-sparing suppressors converge *faster* — lower refinement depth — than damaging perturbations (Cliff's $\delta = -0.16$, $p = 2 \times 10^{-8}$) and than generic "other" perturbations ($\delta = -0.24$, $p = 3 \times 10^{-19}$); they are indistinguishable from inflammatory perturbations on depth alone ($\delta = -0.01$, $p = 0.73$), as expected since both act on the activation axis. The Part-2 hypothesis therefore holds at the stimulated endpoint: suppressors that spare the resting state need less iterative refinement to reach their endpoint than damaging or generic perturbations. Figure 11 summarizes all three results.

Depth clusters and a cell-context-dependent drug negative. Clustering the 12-feature signature ($\mathbb{E}[N]$/halt-confidence/nonlinear/predicted-error $\times$ 3 conditions) yields nine well-separated groups (Leiden modularity 0.66), organized by condition-dependent refinement *dynamics* rather than static pathway family; one cluster is a distinct shallow/fast-converging island ($\mathbb{E}[N] \approx 5.9$ vs 7.2–7.6 elsewhere). Unlike the Replogle lines, where approved and clinical drug targets concentrated in the translation/ribosome depth clusters (Section 5.5, OR 2.8–9.8), no CD4 depth cluster is enriched for drug targets (all OR 0.6–1.3, no significant p): $\approx 97\%$ of the screened CD4 genes are essential, so there is no essentiality contrast to drive the drug/depth co-localization. The drug/depth pattern is itself cell-context-dependent, reinforcing rather than weakening the cell-type-specificity conclusion.

6 Discussion

The positive contribution: a reproducible, effect-independent, cell-type-specific property. The central positive result is a triad. Adaptive-depth halting $\mathbb{E}[N]$, recovered from single-endpoint data on a frozen foundation model, is (i) **reproducible** within a cell type (learned split-half $\rho = 0.80$, below the $\rho \approx 0.97$ effect-size ceiling, so it is stable without being a noiseless copy of effect size); (ii) **effect-size-independent** (partial $\rho \approx 0$ in all four lines once #DE and cell count are controlled, and independent of knockdown strength and target abundance); and (iii) **cell-type-specific** (its sign relative to response magnitude flips between K562 and the other lines, and its ranking does not port across contexts). That such a property exists at all, and is recoverable from the single-endpoint measurements we already collect, is the affirmative claim of the paper.

The negatives are findings, not failures. We regard the negative results as the more durable contribution, and we state them without hedging.

- **Learned halting is not identifiable from single endpoints without scaffolding.** A free learned gate over an expressive predictor collapses to a per-seed constant; anchoring the halt loss to magnitude makes depth reproducible but redundant; removing magnitude makes it non-redundant but irreproducible. A per-perturbation learned depth exists only with a magnitude-free target, a per-perturbation probe token, an unsaturated gate, and oracle supervision derived from the model's own convergence. "Adaptive-depth halting works" is true *only* with this oracle scaffold.
- **The signature is cell-type-specific — it does not port across contexts.** Within a line the depth signature is reproducible (head-seed $\rho = 0.76$–0.85), but cross-line agreement is $\rho = 0.14$ (model-free ceiling 0.011) and cross-donor agreement is $\rho = 0.06$. Depth is a context-specific property of the perturbation, reproducible within a cell type and not shared between them; the three-tier decay $0.93 \to 0.29 \to 0.06$ (head-seed $\to$ from-scratch backbone-seed $\to$ donor) locates the residual model-fit dependence in the from-scratch retrain, not in the deployed fixed-backbone regime.
- **The signature does not sharpen drug-target discovery.** In three of four lines, $\mathbb{E}[N]$ adds nothing to response+STRING similarity for target-class identification; response complexity is largely orthogonal to pharmacological target class.
- **The signature is non-redundant with network topology.** No model-free graph statistic — directed-GRN cascade depth, out-degree, downstream reach, or undirected PPI degree — reproduces the per-perturbation ordering of $\mathbb{E}[N]$ (best correlate $|\rho| = 0.23$, all below a 0.3 novelty ceiling; Section 5.7). Halting depth is therefore not an expensive recomputation of wiring the field already has: it is a genuinely new per-perturbation descriptor, even though it does not port across contexts.

The drug-discovery verdict tallies four independent tests, of which three are negative and the fourth (RPE1) is a weak, drug-class-specific tie-breaker — so the overall application verdict is negative. Under the benchmark-critique framing of this subfield, a clean negative that survives two annotation versions and gene-disjoint cross-validation is a result worth reporting.

Why a frozen foundation model, and what that costs. The identifiability wall fell only when the predictor was a large model pretrained on a genome-wide perturbation atlas, whose layerwise representation of each perturbation is shaped by data far beyond a single endpoint. That is also the sharpest limitation: the signature is a property of *that* pretrained model on *its native* data. It does not transfer to datasets the model cannot predict (e.g. Adamson), and — as the cross-line and cross-donor results show — it does not transfer across cellular contexts even within the model's own training distribution. The claim is "a well-fitted foundation model's layer-exit depth carries a reproducible, effect-independent, context-specific signature," not "this depth is a dataset- or context-universal property of a perturbation."

Limitations. Several caveats bound the interpretation. (1) *Two depth signals.* The computed argmin depth ($\rho = 0.836$) is a readout of the frozen model; the learned $\mathbb{E}[N]$ ($\rho = 0.761$) is a genuine trained gate but requires the oracle scaffold. We report both and never conflate them. (2) *Residual effect coupling.* The learned head carries slightly more effect leakage (raw $\rho = -0.35$) than the computed readout (-0.29); we report the partial correlation, which is ≈ 0. (3) *The oracle τ is a knob.* r^* depends on $\tau = 0.05$; a τ-sensitivity sweep would strengthen the claim, though τ was never tuned on test. (4) *Partial reconstruction fidelity.* Our per-layer decode reaches ≈ 0.67 correlation with ARC's full decode; only the depth *ordering* is used, and it reproduces. (5) *Interpretation ceiling.* Because the data are single endpoints, depth is a computational proxy for response complexity, and no biological-time or causal-depth reading is licensed. (6) *Annotation provenance.* The drug-discovery annotation is Claude-generated from public databases, not expert-curated; candidates are hypotheses.

Future work. The natural next steps are a τ-sensitivity sweep; a same-gene guide-reproducibility test now that the anti-circularity guardrail keeps it non-circular; testing whether the signature is backbone-invariant across ARC's generative and transformer heads; and extending the CD4 single-cell result to more donors to map exactly how cross-donor portability grows along the activation trajectory (the resting-to-48 h transition from $\rho \approx 0$ to $\rho = 0.49$). Whether adaptive depth on a generative backbone (scDiff/CellFlow), whose step-count semantics are cleaner than a transformer's layer axis, yields a portable signature remains open — though the cross-context negatives here suggest portability is a property of the biology, not of the halting mechanism.

7 Conclusion

Adaptive-depth halting $\mathbb{E}[N]$ is a reproducible (within-cell-type split-half $\rho = 0.80$, below the $\rho \approx 0.97$ effect-size ceiling), effect-size-independent, cell-type-specific property of a perturbation's response — recoverable from single-endpoint Perturb-seq — but it is *not* portable across cell types ($\rho = 0.14$) or donors ($\rho = 0.06$), and it does *not* sharpen drug-target-class identification beyond response and network similarity (three of four lines negative). Its sign relative to response magnitude is set by the cellular context. Learned per-perturbation halting is unidentifiable from single endpoints without oracle scaffolding; once that scaffolding is in place on a frozen foundation model, the depth signature is real, but its reach is bounded to the model and context that produced it. The reproducibility, effect-independence, and cell-type-specificity triad is the positive contribution; the identifiability wall, the cross-context non-portability, and the drug-discovery negative are findings in their own right.

Data, code, and reproducibility. All results derive from ARC Institute's public pretrained STATE checkpoints (`ST-SE-Replogle`, four fewshot cell lines) on their native Replogle evaluation data, and from GWCD4i CD4$^+$ CRISPRi Perturb-seq. The model (`adaptive_state_refine.py`), halt-head calibration trainer, per-perturbation signature tables (968–1084 perturbations per line), cross-line and donor-stability data, biological-validation summary, and the drug-discovery bundle (per-line candidate lists, shared annotation over 1708 genes, cross-line consistency) are provided as artifacts. Every reported number is computed from these tables.

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